

Nama : Derren Roosevelt Wijaya

NPM : 252310029

Kelas : TI – 25 – PA2

Tugas 8

Link Github : https://github.com/wijayaderren2-prog/232510029_LabAlgoritma.git

```
1  #include <iostream>
2  #include <string>
3
4  using namespace std;
5
6  const int MAX = 10;
7
8  class Queue {
9  private:
10     string data[MAX];
11     int front;
12     int rear;
13
14 public:
15     Queue() {
16         front = -1;
17         rear = -1;
18     }
19
20     bool isEmpty() {
21         return front == -1;
22     }
23
24     bool isFull() {
25         return rear == MAX - 1;
26     }
27
28     void enqueue(string nama) {
29         if (isFull()) {
30             cout << "Antrian penuh!\n";
31         } else {
32             if (isEmpty()) {
33                 front = 0;
34             }
35             rear++;
36             data[rear] = nama;
37             cout << nama << " berhasil masuk ke antrian.\n";
38         }
39     }
40
41     void dequeue() {
42         if (isEmpty()) {
43             cout << "Antrian kosong!\n";
44         } else {
45             cout << data[front] << " selesai dilayani.\n";
46             if (front == rear) {
47                 front = -1;
48                 rear = -1;
49             }
50         }
51     }
52 }
```

```

49         } else {
50             front++;
51         }
52     }
53 }
54
55 void peek() {
56     if (isEmpty()) {
57         cout << "Antrian kosong.\n";
58     } else {
59         cout << "Antrian terdepan : " << data[front] << endl;
60     }
61 }
62
63 void size() {
64     if (isEmpty()) {
65         cout << "Jumlah antrian : 0\n";
66     } else {
67         cout << "Jumlah antrian : " << rear - front + 1 << endl;
68     }
69 }
70
71 void display() {
72     if (isEmpty()) {
73         cout << "Antrian kosong.\n";
74     } else {
75         cout << "\nDaftar Antrian\n";
76         for (int i = front; i <= rear; i++) {
77             cout << i - front + 1 << ". " << data[i] << endl;
78         }
79     }
80 }
81 };
82
83 int main() {
84     Queue q;
85
86     int pilih;
87     string nama;
88
89     do {
90
91         cout << "\n===== SISTEM ANTREAN LAYANAN PUBLIK =====\n";
92         cout << "1. Ambil Nomor Antrean\n";
93         cout << "2. Panggil Antrean\n";
94         cout << "3. Lihat Antrean Terdepan\n";
95         cout << "4. Jumlah Antrean\n";

```

```

96         cout << "4. Jumlah Antrean\n";
97         cout << "5. Tampilkan Semua Antrean\n";
98         cout << "6. Cek Antrian Kosong\n";
99         cout << "7. Cek Antrian Penuh\n";
100         cout << "8. Keluar\n";
101         cout << "Pilih : ";
102         cin >> pilih;
103         cin.ignore();
104
105         switch(pilih){
106
107             case 1:
108                 cout << "Masukkan Nama : ";
109                 getline(cin,nama);
110                 q.enqueue(nama);
111                 break;
112
113             case 2:
114                 q.dequeue();
115                 break;
116
117             case 3:
118                 q.peek();
119                 break;
120
121             case 4:
122                 q.size();
123                 break;
124
125             case 5:
126                 q.display();
127                 break;
128
129             case 6:
130                 if(q.isEmpty())
131                     cout<<"Antrian kosong.\n";
132                 else
133                     cout<<"Masih ada antrian.\n";
134                 break;
135
136             case 7:
137                 if(q.isFull())
138                     cout<<"Antrian penuh.\n";
139                 else
140                     cout<<"Antrian belum penuh.\n";
141                 break;
142
143             case 8:
144                 cout<<"Program selesai.\n";
145                 break;
146
147             default:
148                 cout<<"Pilihan tidak tersedia.\n";
149         }
150     }while(pilih!=0);
151
152     return 0;
153 }
154

```

===== SISTEM ANTREAN LAYANAN PUBLIK =====

1. Ambil Nomor Antrean
2. Panggil Antrean
3. Lihat Antrean Terdepan
4. Jumlah Antrean
5. Tampilkan Semua Antrean
6. Cek Antrian Kosong
7. Cek Antrian Penuh
0. Keluar

Pilih : 1

Masukkan Nama : Derren

Derren berhasil masuk ke antrian.

===== SISTEM ANTREAN LAYANAN PUBLIK =====

1. Ambil Nomor Antrean
2. Panggil Antrean
3. Lihat Antrean Terdepan
4. Jumlah Antrean
5. Tampilkan Semua Antrean
6. Cek Antrian Kosong
7. Cek Antrian Penuh
0. Keluar

Pilih : 8

Pilihan tidak tersedia.